

Edgar Dolores-Tesillos



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Profile

Atmospheric scientist with expertise in extratropical dynamics, blocking, and climate hazards. Experienced in analyzing high-impact weather events and assessing climate risk using large ensemble datasets and high-resolution simulations. Strong programming background (Python, R and Fortran) and high-performance computing skills. Passionate about applying science to real-world risk challenges.

Education

2018–2022

Ph.D. in Meteorology

Freie Universität Berlin, Berlin, Germany

Thesis: *Changes in Extratropical Cyclone Dynamics in the North Atlantic in a Warming Climate*

2015–2018

M. Sc. in Earth Science

UNAM, Mexico City, Mexico

Thesis: *Detection of the impact of soil moisture on convective initiation in Mexico*

2009–2014

B. Sc. in Atmospheric Sciences

University of Veracruz, Veracruz, Mexico

Thesis: *Characteristic thermodynamics profile for a Norte event in the port of Veracruz.*

Research Experience

2023–Present

Postdoctoral Researcher in weather extremes

University of Lausanne, Switzerland

Project: *HEATForecast: Dynamical constraints for the predictability of heat waves in current and future climates*

- Analyzing the heatwave dynamics in high-resolution climate simulations (ICON)
- Evaluating compound events in new generation of kilometre scale events (e.g., dry-warm events)
- Utilized diagnostic tools in Python and leveraged high-performance computing (Alps, CSCS) to analyze large ensemble datasets

2023–2025

Postdoctoral Researcher in atmospheric dynamics

University of Bern, Switzerland

Project: *Representation and Impacts of Atmospheric Blocking in the nextGEMS*

- Assessed biases in blocking frequency, duration, and dynamics in high-resolution climate simulations

(e.g., ICON, IFS)

- Evaluated large-scale circulation patterns influencing renewable energy variability and extreme weather in Europe
- Engaged in interdisciplinary collaborations and workshops focused on climate hazard impacts

2018–2022

Ph.D. Researcher

Freie Universität Berlin, Germany

Project: *Extratropical Cyclone Dynamics in CESM Large Ensemble*

- Investigated shifts in cyclone tracks and changes in wind and precipitation extremes with direct implications for climate risk
- Developed diagnostic, statistical, and visualization tools using Python and Fortran; analyzed large ensemble climate simulations
- Contributed to publications and presented findings at international conferences on storm-related hazards

Selected Publications

- **Dolores-Tesillos, E.**, Martius, O., and Pfahl, S. (2025). Storm-Resolving Models Advance Atmospheric Blocking Simulations and Climate Change Insights. *EGUsphere*, 2025, 1-45. <https://doi.org/10.5194/egusphere-2025-4969>.
- **Dolores-Tesillos, E.**, Martius, O., and Quinting, J. (2025). On the role of moist and dry processes in atmospheric blocking biases in the Euro-Atlantic region in CMIP6. *Weather Clim. Dynam.*, 6(2), 471–487. <https://doi.org/10.5194/wcd-6-471-2025>.
- **Dolores-Tesillos, E.**, Otero, N., and Allen, S. (2024). Projections of standardised energy indices in future climate scenarios. *Environmental Research Letters*. <https://doi.org/10.1088/1748-9326/ad9b3f>.
- **Dolores-Tesillos, E.**, Teubler, F., and Pfahl, S. (2022). Future changes in North Atlantic winter cyclones in CESM-LE – Part 1: Cyclone intensity, potential vorticity anomalies, and horizontal wind speed, *Weather Clim. Dynam.*, 3, 429–448, <https://doi.org/10.5194/wcd-3-429-2022>.
- **Dolores, E.**, Caetano, E., López-Bravo, L. C., and Calheiros, A. (2019). Influence of soil moisture on mesoscale convective initiation in central Mexico. *European Journal of Remote Sensing*, 52(1), 640–652. <https://doi.org/10.1080/22797254.2019.1700397>.

Skills

Programming: Python (advanced), Fortran, Bash, R, MATLAB

Data & Modeling Tools: xarray, netCDF, CDO, NCO, Git, Pandas, Matplotlib, LaTeX

HPC & Data Handling: Levante (DKRZ), SLURM, large ensemble workflows

Languages: English (fluent), Spanish (native), German (intermediate)

Teaching and Mentorship [Selected]

2025–Present: Lecturer: *Climate Risk Assessment* (labs and tutorials), University of Bern – Topics include hazard modeling, risk communication, and uncertainty analysis

2023–2023: TA: *Meteorologie 3* (tutorials), University of Bern

2020–2022: TA: *Synoptic Meteorology* (labs and tutorials), FU Berlin

2024–2025: Co-supervisor of MSc thesis on Mediterranean cyclones, University of Bern

2024–2024: Co-supervisor of BSc thesis on the wintertime eddy-driven jet stream over the North Atlantic, University of Bern

2023–2024: Co-supervisor of MSc thesis on tracking heatwaves in the Northern Hemisphere, University of Bern

Collaborations and Outreach [Selected]

Workshops: Organizer and instructor: nextGEMS "Hazard Hackathon". 14–18 October 2024, Wageningen, The Netherlands.

Engagement: Speaker at GIUB Colloquium on atmospheric blocking representation in climate models (2025)